Numerical Methods: Euler, RK2, RK4 & Stability

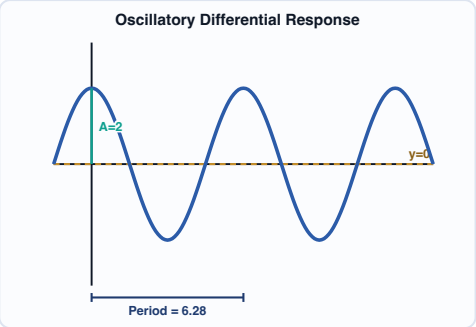
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Numerical Methods: Euler, RK2, RK4 & Stability Practice Exercises

1 MCQ

The second-order linear homogeneous ODE $f''(x) - 39f'(x) + 350f(x) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.



A 39

B 24

C 42

D 53

2 MCQ

Consider the linear first-order differential equation $\frac{dy}{dx} + 2y = 6$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

A 2

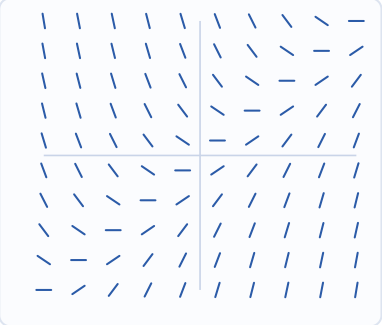
B $e^2 - 1$

C $3 - 2e^{-2}$

D $3 - 2e^{-1}$

3 MCQ

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size of $h = 0.1$ to approximate $y(0.2)$.



A 1.200

B 1.210

C 1.230

D 1.220

4 MCQ

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the auxiliary equation has roots r_1 and r_2, find the value of $r_1 + r_2$.

A 5

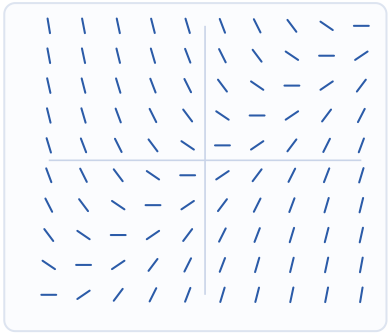
B -5

C 6

D -6

5 MCQ

Consider the separable differential equation $\frac{dy}{dx} = 2x$ with the initial condition $y(0) = 3$. Find the value of $y(2)$.



A 7

B 6

C 8

D 5

6 MCQ

Consider the differential equation $y'' - xy = 0$. The point $x = 0$ is an ordinary point. If we assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$, find the recurrence relation for the coefficients c_n.

A $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$

B $c_{n+2} = \frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

C $c_{n+2} = -\frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$

D $c_{n+2} = -\frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

7 MCQ

The second-order linear homogeneous ODE $u''(s) - 40u'(s) + 144u(s) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

A 44

B 43

C 24

D 40

8 MCQ

The Laplace transform of $u(s) = 170e^{19s}$ is $v(s) = \dfrac{170}{s-19}$ for $s > 19$. Determine the evaluated transform $v(20)$.

A 170

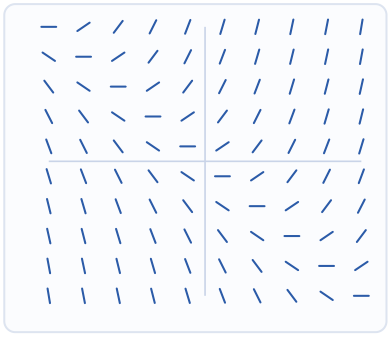
B 151

C 173

D 187

9 MCQ

Consider the differential equation $\dfrac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. You are asked to approximate $y(0.2)$ using Euler's method. If the step size $h = 0.1$ is used, and it is known that the exact solution at $x = 0.2$ is $y(0.2) = 1.24$, determine the value of k such that $y(0.2)_{\text{Euler}} = 1.24 - k$.



A 0.03

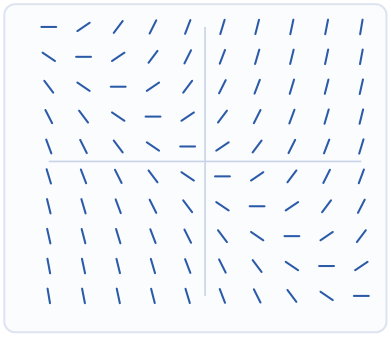
B 0.04

C 0.01

D 0.02

10 MCQ

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size $h = 0.1$ to approximate the value of $y(0.2)$.



A 1.23

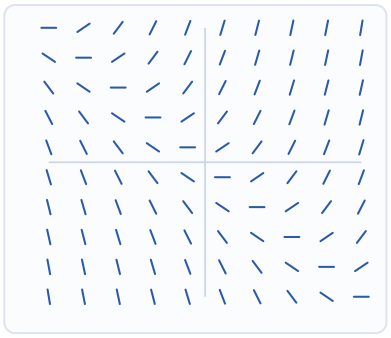
B 1.24

C 1.20

D 1.21

11 MCQ

Consider the ordinary differential equation $\frac{dy}{dx} = x + y$, with the initial condition $y(0) = 1$. We will use Euler's method to approximate the value of $y(0.2)$ using a step size of $h = 0.1$. Let $y_0 = y(x_0)$, $y_1 = y(x_1)$, and $y_2 = y(x_2)$ be the approximations at $x_0 = 0$, $x_1 = 0.1$, and $x_2 = 0.2$ respectively.
Which of the following statements is TRUE regarding these approximations?



A $y_1 = 1.1$ and the approximation for $y(0.2)$ is 1.22

B $y_1 = 1.2$ and the approximation for $y(0.2)$ is 1.32

C $y_1 = 1.2$ and the approximation for $y(0.2)$ is 1.30

D $y_1 = 1.1$ and the approximation for $y(0.2)$ is 1.21

12 MCQ

The first-order differential equation $\frac{df}{dx} = 12x - 2f$ has its slope field plotted below. Determine the slope $\frac{df}{dx}$ at the point $A(6, 23)$.

A 38

B 26

C 3

D 29

13 MCQ

The first-order differential equation $\frac{dp}{dt} = 22t - 5p$ has its slope field plotted below. Determine the slope $\frac{dp}{dt}$ at the point $P(7, 10)$.

A 104

B 94

C 107

D 126

14 MCQ

For Euler's method, dividing the step size h by 10 reduces the global error by roughly what factor?

A 100

B 10

C 10000

D 20

15 MCQ

What is the global truncation error of the improved Euler (Heun) method?

A $O(h^2)$

B $O(h)$

C $O(h^3)$

D $O(h^4)$

16 MCQ

Use Euler's method with $h = \frac{1}{4}$ to approximate $y(\frac{1}{4})$ for $y' = -y$, $y(0) = 3$.

A 3

B $\frac{3}{2}$

C $\frac{75}{32}$

D $\frac{9}{4}$

17 MCQ

For the improved Euler (Heun) method, dividing the step size h by 3 reduces the global error by roughly what factor?

A 3

B 27

C 81

D 9

18 MCQ

Use Euler's method with $h = \frac{1}{4}$ to approximate $y(\frac{1}{4})$ for $y' = 3 - y$, $y(0) = 1$.

A $\frac{3}{2}$

B $\frac{23}{16}$

C 1

D 2

19 MCQ

For the improved Euler (Heun) method, dividing the step size h by 10 reduces the global error by roughly what factor?

A 10

B 1000

C 10000

D 100

20 MCQ

Use Euler's method with $h = \frac{1}{10}$ to approximate $y(\frac{1}{10})$ for $y' = 3 - y$, $y(0) = 1$.

A $\frac{119}{100}$

B $\frac{6}{5}$

C 1

D $\frac{7}{5}$

21 MCQ

Use Euler's method with $h = \frac{1}{5}$ to approximate $y(\frac{6}{5})$ for $y' = 1 + y$, $y(1) = 3$.

A $\frac{97}{25}$

B $\frac{19}{5}$

C 3

D $\frac{23}{5}$

22 MCQ

For Euler's method, dividing the step size h by 3 reduces the global error by roughly what factor?

A 3

B 6

C 9

D 81

23 MCQ

Use Euler's method with $h = \frac{1}{5}$ to approximate $y(\frac{6}{5})$ for $y' = x$, $y(1) = 1$.

A 1

B $\frac{7}{5}$

C $\frac{61}{50}$

D $\frac{6}{5}$

24 MCQ

Use Euler's method with $h = \frac{1}{2}$ to approximate $y(\frac{3}{2})$ for $y' = x^2$, $y(1) = 2$.

A $\frac{45}{16}$

B $\frac{5}{2}$

C 2

D 3

25 MCQ

Use Euler's method with $h = \frac{1}{2}$ to approximate $y(\frac{1}{2})$ for $y' = -y$, $y(0) = 2$.

A 0

B $\frac{5}{4}$

C 2

D 1